

Bachelor of Science in Computer Science University of Colombo School of Computing

SCS 1302 – Discrete Mathematics 1

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Discrete Mathematics

Main Topics :

1. Propositional Logic
2. Predicate Logic
3. Arguments
4. Introduction to Proofs
5. Sets
6. Functions
7. Relation
8. Number Theory

Textbook

Discrete Mathematics and Its Applications

By Kenneth H. Rosen

Published by McGraw-Hill

(7th Edition)

Chapters 1, 2, 4, and 9.

Lesson 1: Propositional Logic

- **Topics to be covered.**
 - **Propositions**
 - **Primitive propositions, logical operators, Compound Propositions and Truth Tables**
 - **Equivalences**
 - **Precedence of Logical Operators**
 - **Tautology and Contradiction**
 - **Normal Forms**
 - **Applications of Propositional Logic**
 - **Satisfiability**

Propositions

A **proposition** is a **declarative sentence** (a sentence that declares a fact) that is either **true (1)** or **false(0)**, but not both. The **true** or **false** value that can be assigned to a proposition is called its **truth value**.

A proposition can be viewed as a sentence that can be assigned with a truth value.

The area of logic that deals with propositions is called the **propositional calculus or propositional logic**. It was first developed systematically by the Greek philosopher Aristotle more than 2300 years ago.

Examples

Examples of propositions:

1. Colombo is the biggest city of Sri Lanka.
2. $2 + 3 = 4$.
3. The Moon is made of cheese.

The following sentences are not propositions:

1. What time is it?
2. Please be silent!
3. $X + 5 = 3$.

True or False of the last statement depends on the value of X . Without knowing it, this statement cannot be assigned with a truth value. The first and second statements are not even declarative statements (facts)

Atomic proposition (Primitive proposition).

A statement that cannot be broken down into smaller statements, is known as an **atomic (primitive)** proposition.

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A statement that cannot be broken down into smaller statements, is known as an **atomic (primitive)** proposition.

Examples

1. 2 is a prime number.
2. $6 > 3$.
3. $3 > 6$.

- Sentence 1 and sentence 2 are true. Sentence 3 is false. These sentences are atomic statements (propositions).

Propositional Variables

- Letters such as p, q, r and s are used to denote atomic propositions. These letters are known as **propositional variables**.
- Generally, a truth value of T (or 1) for true or F (or 0) for false is assigned to each propositional variable.

Composite (or compound) propositions

- A statement formed by combining two or more atomic(simple) propositions using logical connectives such as "and," "or," "not," or "if...then".

Examples:

The sun is hot and water is a liquid.

If it is sunny outside, then I walk to work.

Logical operators (connectors)

- Compound statements are formed by combining more than one primitive proposition by using **logical operators (connectors)**.

Logical Operator (Connector)	Symbol
Negation	$\neg$
Conjunction (and)	$\wedge$
Disjunction (or)	$\vee$
Exclusive Disjunction	$\oplus$
Implication (Conditional)	$\rightarrow$
implication (Bi – conditional)	$\leftrightarrow$

Conjunction – and, product

Disjunction – or , sum

Compound Propositions using propositional variables

The following statements are **compound propositions**?

1. 5 is a prime number **and** $1 = 0$.

p

q

$p \wedge q$

1. Nimal is holding a pencil **or** water is a liquid.

r

s

$r \vee s$

1. **If** Nimal has a dog, **then** cat have 5 lungs.

p

q

$p \rightarrow q$

Compound Propositions - Examples

If compound propositions are also propositions, they should have truth values.

How to determine the truth values of compound positions?

Truth Tables

- Rules to determine the truth values of compound propositions can be given by using truth tables.
- A truth table is a table that displays all possible truth values for the primitive propositions of a compound proposition and the corresponding truth value of the compound proposition.

Negation

Let p be a proposition. The **negation of p** , denoted by $\neg p$, is the statement “It is not the case that p .”

Negation is a unary operator

Example: Let p be “Nimal’s phone has at least 32GB of memory”.

The negation of p is

$\neg$ (Nimal’s phone has at least 32GB of memory)

“It is not the case that Nimal’s phone has at least 32GB of memory.” Or

“Nimal’s phone does not have at least 32GB of memory” Or Simply

“Nimal’s phone has more than 32GB of memory.”

Truth Table for negation:

p	$\neg p$
F	T
T	F

Conjunction & Disjunction

Let p and q be propositions. The **conjunction(product)** of p and q , denoted by $p \wedge q$, is the proposition “ p and q .” The conjunction $p \wedge q$ is true when both p and q are true and is false otherwise.

Let p and q be propositions. The **disjunction(sum)** of p and q , denoted by $p \vee q$, is the proposition “ p or q .” The disjunction $p \vee q$ is false when both p and q are false and is true otherwise.

Truth Table for Conjunction

p	q	$p \wedge q$
F	F	F
F	T	F
T	F	F
T	T	T

Truth Table for Disjunction

p	q	$p \vee q$
F	F	F
F	T	T
T	F	T
T	T	T

Conjunctions

- Note that in logic the word “but” sometimes is used instead of “and” in a conjunction.
- “The sun is shining, but it is raining” is another way of saying “The sun is shining and it is raining.”

Examples

Let p and q represent the propositions:

p : It is below freezing. q : It is snowing.

Write these propositions using p and q and logical connectives (including negations).

- It is below freezing and snowing.

$$p \wedge q$$

- It is below freezing and not snowing.

$$p \wedge \neg q$$

- It is not below freezing, and it is not snowing.

$$\neg p \wedge \neg q$$

- It is either snowing or below freezing (**or both**).

$$p \vee q$$

Exclusive Disjunction (exclusive or)

Let p and q be propositions. The **exclusive or** of p and q , denoted by $p \oplus q$, is the proposition that is true when exactly one of p and q is true and is false otherwise.

Example: “ Either soup or salad comes with the meal”

The truth table for the exclusive or

p	q	$p \oplus q$
F	F	F
F	T	T
T	F	T
T	T	F

Conditional Statement or Implication

Let p and q be propositions. The **conditional statement** $p \rightarrow q$ is the proposition “**if p , then q .**” The conditional statement $p \rightarrow q$ is false when p is true and q is false, and true otherwise.

p is called the **hypothesis** (or antecedent or premise) and q is called the **conclusion** (or consequence).

Various terminology used:

- “ p is sufficient for q ”
- “a sufficient condition for q is p ”
- p only if q
- “ p implies q ”
- “ q whenever p ”

Truth Table for $p \rightarrow q$

p	q	$p \rightarrow q$
F	F	T
F	T	T
T	F	F
T	T	T

Example

- Consider the following statement:

If it rains, It will be cloudy

x

y

$x \rightarrow y$

- Case 1: It rained then we say, It should be cloudy. So If x happened then y should happen. $x \rightarrow y$
- Case 2: There are no clouds, So there is no rain. $\sim y \rightarrow \sim x$
- Case 3: It is not raining. Uncertain. As there may be clouds or may not be.
- Case 4: It is cloudy. Uncertain. As it may rain may not rain.

Example

p	q	$P \rightarrow q$
F	F	T
F	T	T
T	F	F
T	T	T

If it rains, It will be cloudy

Example

p	q	$P \rightarrow q$
F	F	T
F	T	T
T	F	F
T	T	T

If it rains, It will be cloudy

Example

p	q	$P \rightarrow q$
F	F	T
F	T	T
T	F	F
T	T	T

If it rains, It will be cloudy

Examples of a conditional statement

Let p be “Kamal learns discrete mathematics” and
 q be “Kamal will find a good job.”

Express the statement $p \rightarrow q$ as a statement in English.

Solution: $p \rightarrow q$ represents the proposition “If Kamal learns discrete mathematics, then he will find a good job.”

$p \rightarrow q$ can also be expressed as:

- “Kamal will find a good job when he learns discrete mathematics.”
- “For Kamal to get a good job, it is sufficient for him to learn discrete mathematics.”
- “Kamal will find a good job whenever he learns discrete mathematics.”

Propositions Vs. Formula

- A formula is a preposition where all its propositions are replaced with propositional variables.

Example :

$$p \wedge q$$

Equivalent propositions

When two **compound propositions** always have the same truth values for all possible truth values for its primitive propositions, we call them **logically equivalent**($\equiv$).

$p \rightarrow q$ and $\sim q \rightarrow \sim p$ are equivalent ($p \rightarrow q \equiv \sim q \rightarrow \sim p$)

If two propositions are equivalent one can be replaced with the other without changing the result.

Note: $\equiv$ is not a logical connective and $p \equiv q$ is not a compound proposition.

If $p \equiv q$ implies that $p \leftrightarrow q$ is a tautology

Equivalent propositions

Equivalent of propositions enables compound propositions to be replaced by another compound propositions that are logically equivalent to them without changing the truth value of the original compound propositions.

Equivalence

p	q	$\neg p$	$\neg q$	$p \rightarrow q$	$q \rightarrow p$	$\neg q \rightarrow \neg p$	$\neg p \rightarrow \neg q$
0	0	1	1	1	1	1	1
0	1	1	0	1	0	1	0
1	0	0	1	0	1	0	1
1	1	0	0	1	1	1	1

$$p \rightarrow q \equiv \neg q \rightarrow \neg p$$

$$q \rightarrow p \equiv \neg p \rightarrow \neg q$$

Bi-conditionals or Bi-implications

Let p and q be propositions. The **bi-conditional** statement $p \leftrightarrow q$ is the proposition “ p if and only if q .” The bi-conditional statement $p \leftrightarrow q$ is true when p and q have the same truth values, and is false otherwise.

Truth Table for $p \leftrightarrow q$

p	q	$p \leftrightarrow q$
F	F	T
F	T	F
T	F	F
T	T	T

Other ways to express $p \leftrightarrow q$:

- “ p is necessary and sufficient for q ”
- “if p then q , and if q then p ”
- “ p iff q .”

Example: “You can have dessert if and only if you finish your meal.”

Precedence of Logical Operators

$\vee$ and $\rightarrow$ are binary operators

Does $p \vee q \rightarrow r$ means

1) $(p \vee q) \rightarrow r$ or

2) $p \vee (q \rightarrow r)$

Precedence of Logical Operators

Precedence of Logical Operators	
Operator	Precedence (Highest to lowest)
$\neg$	1
$\wedge$	2
$\vee$	3
$\rightarrow$	4
$\leftrightarrow$	5

$p \vee q \rightarrow r$ is the same as $(p \vee q) \rightarrow r$

$$\neg p \rightarrow q \vee r \rightarrow p \vee q \wedge r \equiv (\neg p) \rightarrow ((q \vee r) \rightarrow [p \vee (q \wedge r)])$$

Precedence of Logical Operators

- All operators are right-associative

$$p \vee q \vee r$$

$$(p \vee (q \vee r))$$

$$5 - 3 + 2$$

$$(5 - 3) + 2 = 4$$

$$5 - (3+2) = 0$$

Construction of truth tables for compound propositions

Let a compound proposition C consist of the atomic propositions $p_1; p_2; \dots; p_n$.

- To compute the truth table do the following:
 1. List its atomic propositions $p_1; p_2; \dots; p_n$ and count their number, n .
 2. Form a table with $m = 2^n$ rows.
 3. List all possible combinations of the truth values for $p_1; p_2; \dots; p_n$.
 4. Prioritize the logical operations in C .
 5. Compute the truth values in order of the operators.

Example

- Let the compound proposition C be $(p \wedge q) \rightarrow p$.
Count the number of atomic propositions, $n = 2$.
Form a table with $m = 2^2 = 4$ rows.
List all possible combinations of the truth values for p and q.
Prioritize the logical operations in the proposition $(p \overset{1}{\wedge} q) \overset{2}{\rightarrow} p$.
- Compute first 1 and then 2

Example

p	q	$p \wedge q$	$(p \wedge q) \rightarrow p.$
T	T		
T	F		
F	T		
F	F		

Example

p	q	$p \wedge q$	$(p \wedge q) \rightarrow p.$
T	T	T	
T	F	F	
F	T	F	
F	F	F	

Example

p	q	$p \wedge q$	$(p \wedge q) \rightarrow p.$
T	T	T	T
T	F	F	T
F	T	F	T
F	F	F	T

p	q	$p \rightarrow q$
F	F	T
F	T	T
T	F	F
T	T	T

Truth Tables of Compound Propositions

Construct the truth table of the compound proposition
 $(p \vee \neg q) \rightarrow (p \wedge q).$

p	q	$\neg q$	$p \vee \neg q$	$p \wedge q$	$(p \vee \neg q) \rightarrow (p \wedge q)$
0	0	1	1	0	0
0	1	0	0	0	1
1	0	1	1	0	0
1	1	0	1	1	1

Construction of truth tables

p	q	r
T		
T		
T		
T		
F		
F		
F		
F		

Construction of truth tables

p	q	r
T	T	
T	T	
T	F	
T	F	
F	T	
F	T	
F	F	
F	F	

Construction of truth tables

p	q	r
T	T	T
T	T	F
T	F	T
T	F	F
F	T	T
F	T	F
F	F	T
F	F	F

Tautology and Contradiction

A compound proposition that is always true, no matter what the truth values of the propositional variables that occur in it, is called a **tautology**. A compound proposition that is always false is called a **contradiction**. A compound proposition that is neither a tautology nor a contradiction is called a **contingency**.

Tautology		
p	$\neg p$	$p \vee \neg p$
T	F	T
F	T	T

Contradiction		
p	$\neg p$	$p \wedge \neg p$
T	F	F
F	T	F

If $p \equiv q$ implies that $p \leftrightarrow q$ is a tautology

$$p \rightarrow q \equiv \neg p \vee q$$

p	q	$p \rightarrow q$	$\neg p$	$\neg p \vee q$	$(p \rightarrow q) \leftrightarrow (\neg p \vee q)$
0	0	1	1	1	1
0	1	1	1	1	1
1	0	0	0	0	1
1	1	1	0	1	1

Logical Equivalences

Identity laws:

$$p \wedge T \equiv p$$
$$p \vee F \equiv p$$

Domination laws:

$$p \vee T \equiv T$$
$$p \wedge F \equiv F$$

Idempotent laws:

$$p \vee p \equiv p$$
$$p \wedge p \equiv p$$

Commutative laws:

$$p \vee q \equiv q \vee p$$
$$p \wedge q \equiv q \wedge p$$

Associative laws:

$$(p \vee q) \vee r \equiv p \vee (q \vee r)$$
$$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$$

Distributive laws:

$$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$$
$$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$$

De Morgan's laws:

$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$
$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$

Absorption laws:

$$p \vee (p \wedge q) \equiv p$$
$$p \wedge (p \vee q) \equiv p$$

Negation laws:

$$p \vee \neg p \equiv T$$
$$p \wedge \neg p \equiv F$$

Double Negation law:

$$\neg(\neg p) \equiv p.$$

Logical Equivalences

$$P \oplus q \equiv (p \wedge \neg q) \vee (\neg p \wedge q)$$
$$P \oplus q \equiv \neg(p \wedge q) \wedge (p \vee q)$$

p	q	$\neg p$	$\neg q$	$p \wedge \neg q$	$(p \wedge \neg q)$	$(\neg p \wedge q)$	$(p \wedge \neg q) \vee (\neg p \wedge q)$	$p \oplus q$
T	T	F	F	F	F	F	F	F
T	F	F	T	T	T	F	T	T
F	T	T	F	F	F	T	T	T
F	F	T	T	F	F	F	F	F

Principle of Duality

- The dual of a proposition is formed by interchanging "AND" with "OR" operations, and 0 with 1 (representing false and true, respectively).
- The principle of duality in logic states that if a proposition is true, its dual is also true.

Examples

Identity Law: $p \wedge T \equiv p$

p	T	$p \wedge T$
F	T	F
T	T	T

De Morgan's Law: $\neg(p \wedge q) \equiv \neg p \vee \neg q$

p	q	$\neg p$	$\neg q$	$(p \wedge q)$	$\neg(p \wedge q)$	$\neg p \vee \neg q$
0	0	1	1	0	1	1
0	1	1	0	0	1	1
1	0	0	1	0	1	1
1	1	0	0	1	0	0

Distributive Law:

$$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$$

p	q	r	$(q \wedge r)$	$p \vee (q \wedge r)$	$(p \vee q)$	$(p \vee r)$	$(p \vee q) \wedge (p \vee r)$
0	0	0	0	0	0	0	0
0	0	1	0	0	0	1	0
0	1	0	0	0	1	0	0
0	1	1	1	1	1	1	1
1	0	0	0	1	1	1	1
1	0	1	0	1	1	1	1
1	1	0	0	1	1	1	1
1	1	1	1	1	1	1	1

More Logical Equivalences

Logical Equivalences Involving Conditional Statements

$$p \rightarrow q \equiv \neg p \vee q$$

$$p \rightarrow q \equiv \neg q \rightarrow \neg p$$

$$p \vee q \equiv \neg p \rightarrow q$$

$$p \wedge q \equiv \neg(p \rightarrow \neg q)$$

$$\neg(p \rightarrow q) \equiv p \wedge \neg q$$

$$(p \rightarrow q) \wedge (p \rightarrow r) \equiv p \rightarrow (q \wedge r)$$

$$(p \rightarrow r) \wedge (q \rightarrow r) \equiv (p \vee q) \rightarrow r$$

$$(p \rightarrow q) \vee (p \rightarrow r) \equiv p \rightarrow (q \vee r)$$

Logical Equivalences Involving Bi-conditional Statements

$$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$$

$$p \leftrightarrow q \equiv \neg p \leftrightarrow \neg q$$

$$p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$$

$$\neg(p \leftrightarrow q) \equiv p \leftrightarrow \neg q$$

Examples

Prove the following logical identities by developing a series of logical equivalences:

1. $\neg(p \vee (\neg p \wedge q)) \equiv \neg p \wedge \neg q$
2. $(p \wedge q) \rightarrow (p \vee q) \equiv T$
3. $(p \rightarrow q) \vee (p \rightarrow r) \equiv \neg p \vee q \vee r.$

$$\begin{aligned}\neg(p \vee (\neg p \wedge q)) &\equiv \neg p \wedge \neg(\neg p \wedge q) && (\because \text{De Morgan's Law}) \\ &\equiv \neg p \wedge [\neg(\neg p) \vee \neg q] && (\because \text{De Morgan's Law}) \\ &\equiv \neg p \wedge [p \vee \neg q] && (\because \text{Double negation}) \\ &\equiv (\neg p \wedge p) \vee (\neg p \wedge \neg q) && (\because \text{Distributive law}) \\ &\equiv F \vee (\neg p \wedge \neg q) && (\because \text{Negation law}) \\ &\equiv \neg p \wedge \neg q && (\because \text{Identity law})\end{aligned}$$

Do we need all logical operators?

$\vee$ - Disjunction

$\wedge$ - Conjunction $p \wedge q \equiv \neg(\neg p \vee \neg q)$.

$\neg$ - Negation

$\rightarrow$ - Implication $p \rightarrow q \equiv \neg p \vee q$

$\oplus$ - Exclusive or $p \oplus q \equiv (p \wedge \neg q) \vee (\neg p \wedge q)$

$\leftrightarrow$ - Bi-conditional $p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$
 $\equiv (\neg p \vee q) \wedge (\neg q \vee p)$.

Functionally Complete

A set of logical operators is called **functionally complete** if every compound proposition is logically equivalent to a compound proposition involving only this set of logical operators.

The set $\{\vee, \neg\}$ form a functionally complete set of operators.

Normal Forms

A product (conjunction) of the variables and their negations in a proposition is called an **elementary product**. E.g: $p \wedge q \wedge \neg r$

A sum (disjunction) of the variables and their negations in a proposition is called an **elementary sum**. Example: $p \vee \neg q \vee r$

Disjunctive Normal Form: A proposition which is equivalent to a proposition which consists of a sum($\vee$) of elementary products ($\wedge$) is called a disjunctive normal form of the given formula.

Example: $(p \wedge q) \vee (p \wedge \neg q) \vee (\neg p \wedge q)$

Conjunctive Normal Form: A proposition which is equivalent to a given proposition which consists of a product of elementary sums is called a conjunctive normal form of the given formula.

Example : $(p \vee q \vee r) \wedge (\neg p \vee q \vee r)$.

Normal Forms

If a compound proposition comprises of only conjunction and negation operators it is said to be in **elementary product form**.

E.g: $p \wedge q \wedge \neg r$

IF A sum (disjunction) of the variables and their negations in a formula is called an **elementary sum form**. Example: $p \vee \neg q \vee r$

Disjunctive Normal Form: A proposition which is equivalent to a given proposition, and which consists of a sum of elementary products is called a disjunctive normal form of the given formula.

Example: $(p \wedge q) \vee (p \wedge \neg q) \vee (\neg p \wedge q)$

Normal Forms

Conjunctive Normal Form: A proposition which is equivalent to a given proposition, and which consists of a product of elementary sums is called a conjunctive normal form of the given proposition.

Example : $(p \vee q \vee r) \wedge (\neg p \vee q \vee r)$.

Example -1 (method 1)

Find the disjunctive normal form of the proposition $(\neg p \vee \neg q) \rightarrow (p \leftrightarrow \neg q)$.

$$\begin{aligned}(\neg p \vee \neg q) \rightarrow (p \leftrightarrow \neg q) &\equiv \neg(\neg p \vee \neg q) \vee (p \leftrightarrow \neg q) \\&\equiv (p \wedge q) \vee \left[(p \wedge \neg q) \vee (\neg p \wedge (\neg(\neg q))) \right] \\&\equiv (p \wedge q) \vee [(p \wedge \neg q) \vee (\neg p \wedge q)] \\&\equiv (p \wedge q) \vee (p \wedge \neg q) \vee (\neg p \wedge q).\end{aligned}$$

Example -1 (method 2)

Find the disjunctive normal form of the formula
 $x = (\neg p \vee \neg q) \rightarrow (p \leftrightarrow \neg q)$.

p	q	$\neg p$	$\neg q$	$(\neg p \vee \neg q)$	$(p \leftrightarrow \neg q)$	x
0	0	1	1	1	0	0
0	1	1	0	1	1	1
1	0	0	1	1	1	1
1	1	0	0	0	0	1

$$x = (\neg p \wedge q) \vee (p \wedge \neg q) \vee (p \wedge q).$$

Example -2 (method -1)

Find the conjunctive normal form of the formula
 $x = (\neg p \vee \neg q) \rightarrow (p \leftrightarrow \neg q)$.

$$\begin{aligned} & (\neg p \vee \neg q) \rightarrow (p \leftrightarrow \neg q) \\ \equiv & \neg(\neg p \vee \neg q) \vee (p \leftrightarrow \neg q) \\ \equiv & (p \wedge q) \vee [(p \rightarrow \neg q) \wedge (\neg q \rightarrow p)] \\ \equiv & (p \wedge q) \vee [(\neg p \vee \neg q) \wedge (q \vee p)] \\ \equiv & [p \vee ((\neg p \vee \neg q) \wedge (q \vee p))] \wedge [q \vee ((\neg p \vee \neg q) \wedge (q \vee p))] \\ \equiv & [(p \vee \neg p \vee \neg q) \wedge (p \vee q \vee p)] \wedge [(q \vee \neg p \vee \neg q) \wedge (q \vee q \vee p)] \\ \equiv & [T \wedge (p \vee q)] \wedge [T \wedge (p \vee q)] \\ \equiv & (p \vee q). \end{aligned}$$

Example (method 2)

Find the conjunctive normal form of the formula
 $x = (\neg p \vee \neg q) \rightarrow (p \leftrightarrow \neg q)$.

p	q	$\neg p$	$\neg q$	$(\neg p \vee \neg q)$	$(p \leftrightarrow \neg q)$	x	$\neg x$
0	0	1	1	1	0	0	1
0	1	1	0	1	1	1	0
1	0	0	1	1	1	1	0
1	1	0	0	0	0	1	0

$$\neg x = (\neg p \wedge \neg q)$$

$$x = \neg(\neg x) = \neg(\neg p \wedge \neg q) = (p \vee q).$$

Applications of Propositional Logic

- Translating English sentences into expressions involving propositional variables and logical connectives.
- Translating Logical Expressions into English Sentences
- System specifications
- Logic circuits
- Logic puzzles

Translating English Sentences

Translate the following English sentences into a logical expression:

“You can access the Internet from campus only if you are a computer science major or you are not a freshman.”

Solution:

Let a := “You can access the Internet from campus”

c := “You are a computer science major”, and

f := “You are a freshman”.

Given sentence can be represented as $a \rightarrow (c \vee \neg f)$.

Note: *The conditional statement $p \rightarrow q$ can also be expressed as “ p only if q ”.*

Exercise

Translate the following English sentences into a logical expression:

- “School is closed if more than 2 feet of snow falls or if the wind chill is below -100.”
- “To take discrete mathematics, you must have taken calculus or a course in computer science.”

Translating Logical Expressions into English Sentences

Let P be “It is snowing”,
 q be “I will go to town”, and
 r be “I have time”.

Write a sentence in English corresponding to each of the following propositions:

1. $r \wedge q$
2. $q \leftrightarrow (r \wedge \neg p)$
3. $(q \rightarrow r) \wedge (r \rightarrow q)$
4. $\neg(q \vee r)$.

Classification of Propositions – Satisfiable and Unsatisfiable

A formula (proposition) is **satisfiable** if there is an assignment of truth values to its variables (propositions) that makes it true (that is, when it is a tautology or a contingency)

Examples

$$p, \neg p, p \wedge q$$

In Logic a PROPOSITION C is called UNSATISFIABLE if there are no output values T (true) in the truth table for C.

Examples

- $p \wedge \neg p$
- when the compound proposition is false for all assignments of truth values to its variables, the compound proposition is **unsatisfiable**

Satisfiable and Unsatisfiable

- To show that a compound proposition is unsatisfiable need to show that *every* assignment of truth values to its variables makes it false.

Classification of Propositions – Valid

In Logic a proposition C is called VALID if its output values in the truth table are all T (true)-

Example;

$$p \vee \neg p$$

Logical Consequence

A proposition B is a logical consequence of a collection of propositions $A_1; A_2; A_3; \dots; A_n$ if the proposition $(A_1 \wedge A_2 \wedge A_3 \wedge \dots \wedge A_n \rightarrow B)$ is valid

Example:

From p and $p \rightarrow q$ can we conclude q ?

Example

p	q	$p \rightarrow q$	$p \wedge (p \rightarrow q)$	$(p \wedge (p \rightarrow q)) \rightarrow q$
T	T	T	T	T
T	F	F	F	T
F	T	T	F	T
F	F	T	F	T

Since $(p \wedge (p \rightarrow q)) \rightarrow q$ is a valid proposition q can be deduced from the propositions p and $p \rightarrow q$

Consistency of propositions

- A set of propositions are consistent if and only if there exists an assignment of truth values for *all its propositional variables* that makes all propositions are true at the same time.

System Specifications - Example

Determine whether these system specifications are consistent:

- “The diagnostic message is stored in the buffer or it is retransmitted.”
- “The diagnostic message is not stored in the buffer.”
- “If the diagnostic message is stored in the buffer, then it is retransmitted.”

Let p denote “The diagnostic message is stored in the buffer”, and

let q denote “The diagnostic message is retransmitted.”

Specifications as logical expressions are $p \vee q$, $\neg p$, and $p \rightarrow q$.

These system specifications are consistent if and only if there exists an assignment of truth values for p and q that makes all three specifications true.

Example Continued

p	q	$\neg p$	$p \vee q$	$p \rightarrow q$	$\neg p \wedge (p \vee q) \wedge (p \rightarrow q)$
0	0	1	0	1	0
0	1	1	1	1	1
1	0	0	1	0	0
1	1	0	1	1	0

When p is false and q is true all the specifications are true. System is consistent.

Method 2: Suppose that $\neg p$ and $p \vee q$ are true. Then

p is false ($\because \neg p$ is true)

$\Rightarrow q$ is true ($\because p \vee q$ is true)

$\Rightarrow p \rightarrow q$ is true

$\Rightarrow$ When p is false and q is true all the specifications are true. System of specifications are consistent.

System Specifications

- System specifications should be consistent, that is, they should not contain conflicting requirements that could be used to derive a contradiction.
- When specifications are not consistent, there would be no way to develop a system that satisfies all specifications.

Example

Determine whether the following system specifications are consistent:

The system is in multiuser state if and only if it is operating normally. If the system is operating normally, then the kernel is functioning. The kernel is not functioning or the system is in interrupt mode. If the system is not in multiuser state, then it is in interrupt mode. The system is not in interrupt mode.

Let p denote “The system is in multiuser state”;

q denote “The system is operating normally”;

r denote “The kernel is functioning” ; and

s denote “The system is in interrupt mode.”

Specifications as logical expressions are $p \leftrightarrow q$, $q \rightarrow r$, $\neg r \vee s$, $\neg p \rightarrow s$ and $\neg s$.

Last Example of Topic -1

Specifications as logical expressions are $p \leftrightarrow q$, $q \rightarrow r$, $\neg r \vee s$, $\neg p \rightarrow s$ and $\neg s$.

Suppose that $q \rightarrow r$, $\neg r \vee s$, $\neg p \rightarrow s$ and $\neg s$ are true. Then
 s is false ($\because \neg s$ is true)

Also $\neg p$ is false ($\because \neg p \rightarrow s$ is true) and hence p is true.

Since $\neg r \vee s$ is true and s is false, $\neg r$ must be true and hence r is false.

Since $q \rightarrow r$ is true and r is false, q must be false.

Now p is true and q is false. Therefore, $p \leftrightarrow q$ is false.

That is, whenever $q \rightarrow r$, $\neg r \vee s$, $\neg p \rightarrow s$ and $\neg s$ are all true, $p \leftrightarrow q$ is **not** true.

Therefore, this system of specifications is **not consistent**.

Theorem???

- Once we prove a mathematical statement is true, we call it a theorem.
- Knowing the proof of a theorem often makes it possible to modify the result to fit new situations.